

Adversarial Training

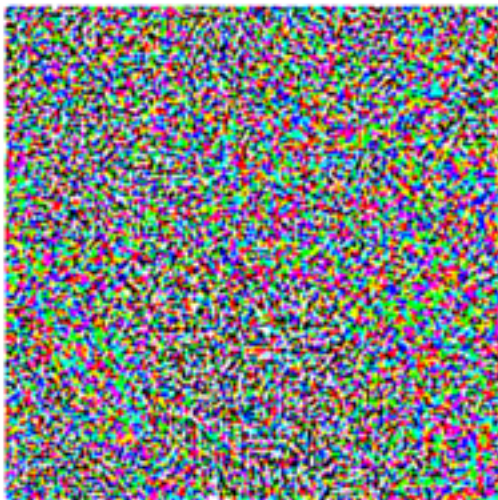
$$\min_f \mathbb{E}_{x,y} \max_{|\delta|_\infty \leq \epsilon} \text{loss}(f(x + \delta), y)$$

Works for classification tasks and small perturbation

Does **not** for LLMs!

1. The set of *all possible* adversarial prompts is too large
2. Many attacks (PAIR, TAP, etc) completely change the prompts!!!

$+ .007 \times$



$=$

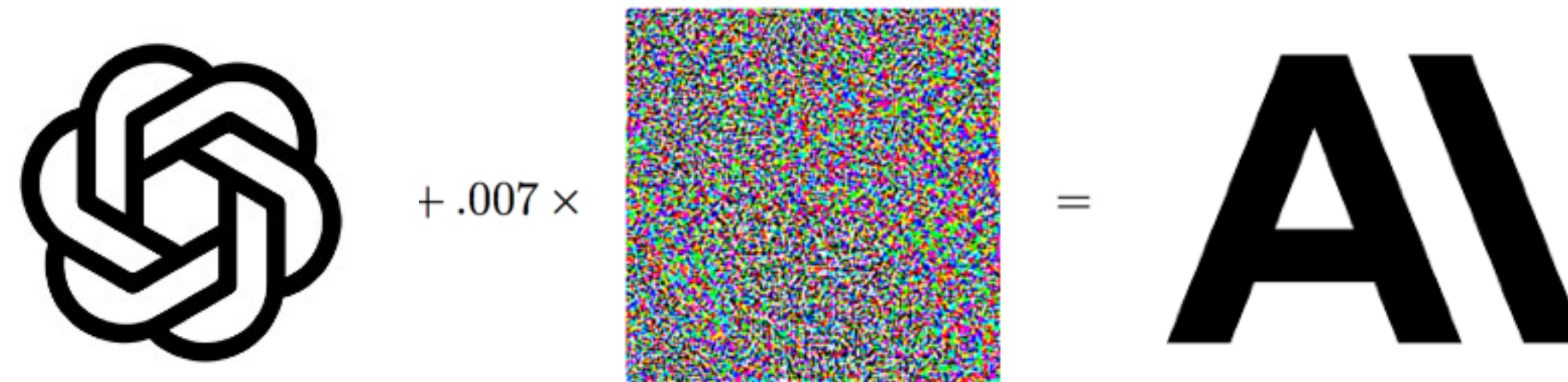


AI

Jailbreaking defenses

Jailbreaking defenses

Adversarial Training



$$\min_f \mathbb{E}_{x,y} \max_{|\delta|_\infty \leq \epsilon} \text{loss}(f(x + \delta), y)$$

Works for classification tasks and small perturbation set

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Swiss cheese model of AI defense

Improving robustness by implementing different **layers of security**.

